CSC 251 HOMEWORK FIVE

STUDENT RECORDS

THE ASSIGNMENT

This assignment will provide practice in using file I/O to read in student records, store the information in an appropriate Python data structure, sort by grade, and output a statistical summary.

PROGRAM DETAILS

The user should be prompted for the name of an input text file. Your program should read records from the file until the end of file is reached. A complete record will contain a student's full name, their ID number, and an exam grade, each on a separate line. You should read until the end of the file, storing the information in an appropriate data structure.

Your program should output a formatted table, sorted by descending exam grade. Assume that a score of 0 indicates the student was not present to take the exam and replace their score with an "N/A". Include a statistical summary table for the non-zero grades, indicating the minimum (should not be 0), maximum, mean, and median.

Your program should handle all exceptions such that it does not crash. You should handle errors in the following way:

1) For I/O errors, output an "Exception thrown:" alert followed by the exception's default error message and prompt the user again until a valid filename is entered so that the program can continue.

2) For value errors, output an "Exception thrown:" alert followed by the exception's default error message then output a table of sorted information and stats based upon the records read in successfully up to that point. This would apply in the case of an incomplete record or invalid input.

You are also expected to manage data for any field when the input is not what is expected. Names that are too long to fit within the 20-character column of the given output table should be truncated to 19 characters, allowing one space between fields for readability. ID numbers that exceed 7 characters should also be truncated to 7 digits. Unreasonable exam scores (below 0 or above 110% in the case of extra credit) should be given a value of N/A and should not be included in calculations.

A student should only appear once in the table output. If multiple records exist for the same student in a datafile, update the student's grade with the second (or last) entry.

REQUIRED FUNCTIONS

Your program should include at least the following functions:

1) A `main` function consisting mainly of calls to other functions and any output statements.

2) A `load_info` function that prompts the user for a filename and reads the records until the end of the file is reached or an exception is thrown. The function should return the data structure used to store the student records.

3) A `print_info` function that is passed the data structure. The function should print out the student info in table form. The title and table heading are provided for you in the starter file called `hw5_table_header.py`. Import this file as a module and use this function to output the main table header. Any columns containing strings should be left-justified while columns with numerical values should be right-justified. Exam grades should be read in and output to the nearest whole number, rounding if the grade is entered as a float. This function is not responsible for outputting the statistical summary.

4) A `sort_info` function that is passed the data structure and sorts the records by **decreasing exam grade**. Be sure to keep the information in a single record together. Students with identical grades should be listed in the order they were entered into the student record.

5) A `get_mean` function that is passed the data structure and returns the mean of the exam values for the exams taken. The mean should be output from `main` to 1 decimal place.

6) A `get_median` function that is passed the data structure and returns the median (middle grade in the sorted list) of the non-zero exam values. The median should be output from `main` to 1 decimal place.

SAMPLE INPUT

The format of the `input02.txt` datafile is:

```
Josh Nelson
1419416
83
Samantha Stone
9209332
95
Hillary Black
8192301
70
Ben Branson
4389210
91
Johnnie Cash
3892012
66
Bradley Long
6472810
0
```

```
Aaron Moore
2020190
100
. . .
```

SAMPLE EXECUTION

Your output should for the `input02.txt` sample file should match the following. Note: user input is shown in bold.

```
Enter input filename: input02.txt

Student Grade Data

Name                       ID   Grade
-----------------------------------
Ginger Vitus          8882384     102
Aaron Moore           2020190     100
Mabel Table           2323232     100
Amelia Bedelia        2939292      98
Samantha Stone        9209332      95
Palek Patel           6467485      95
Gretel Snitzel        7717711      93
Ben Branson           4389210      91
Yin Yang              9399399      89
Wally Pocket          1234567      86
Josh Nelson           1419416      83
Ruthie Toothie        1273841      77
Muffie Huffie         2722734      75
Jason Mamou           1881812      75
Tim Trim              1000101      75
Hillary Black         8192301      70
Johnnie Cash          3892012      66
Paul Small            1000002      65
Tyler Taylor          3838381      55
Wes Beston            1010101      52
Mary Moore            9102934      35
Bradley Long          6472810     N/A
Jack Slack            5757575     N/A

Exam Stats
-----------------------------------
Minimum                            35
Maximum                           102
Mean                             79.9
Median                           83.0
```

PROGRAM SUBMISSION

Submit a single source code file named `HW5LastnameFirstInitial.py` with appropriate documentation for your file and functions. Note: ALL functions should include a docstring-style comment!